Is the volume-law horizon a postulate or a consequence? A reduction to a driven non-equilibrium conjecture for Structural Entropy Dark Energy

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Abstract

Structural Entropy Dark Energy (SEDE) reproduces the cosmological data with no fitted dark-sector parameter, at the price of one foundational input: the cosmic apparent horizon carries volume-law entropy (Barrow exponent $\Delta = 1$) — scaling with the enclosed volume rather than the boundary area. We ask how much of that input is irreducible. The postulate decomposes into *state, form, scale,* and *count*; state, form and scale are first-principles or reducible given the standard identification of the gravitational horizon entropy with an entanglement entropy — with the caveat, stated up front, that a *volume*-law coefficient (unlike the area law's universal $A/4G$) is state-dependent, so *scale* reduces only *given* the Bousso-saturated horizon state that fixes $s_{\rm grav}$: a conditional, not a universal coefficient. That leaves the bulk-versus-boundary count as the residue — which a Bekenstein no-go shows cannot follow from any energy bound. We then analyse the *selection* of the volume branch dynamically, as a motivated conditional rather than a derivation: because gravity's long-range cooperative coupling clears the ordering threshold only with a size-dependent ratio, the two-phase free energy did not exist at the horizon's birth; grown continuously from Planckian size, the horizon passes *through* the ordering bifurcation with Barrow entropy tilting the nascent well toward volume, so $\Delta = 1$ is *favoured* at birth, and long-range hysteresis locks each horizon to its birth branch — the cosmic horizon to volume, an abruptly-formed black hole to area. We are explicit about the limits of this argument: the tie-breaking tilt is itself the Barrow-$\Delta = 1$ free energy, the two-well landscape presupposes an accessible volume phase, the mean-field treatment is extrapolated to the N $\sim$ O(1) quantum-gravity regime where its tools are not controlled, and the one concrete model of de Sitter holography (double-scaled SYK) currently maps to the *area* branch — so the mechanism relocates and narrows the postulate, it does not eliminate it. Its lasting value is to convert an untestable counting postulate into a falsifiable one: a pre-registered point prediction $\Delta = 1$, tested to a forecast $\sigma(\Delta) \approx 0.09$ by DESI DR3 + Euclid. On public data the contrast is already sharp — in a *profile*-likelihood analysis (the marginalised posterior awaits DR3+Euclid) the structure-gated family *prefers* the volume endpoint ($\Delta = 0.93$ [0.83, 1.02]) that the un-gated Barrow family pays $\Delta\chi^2 \approx 2400$ to reach. The marginalised model-level statement is more modest: at equal parameter count the zero-parameter $\Delta = 1$ model is favoured over ΛCDM at the cosmology paper's ΔDIC ≈ -3 (a point-fit gives $\Delta\chi^2 \approx 2.9$, but the DIC is the honest headline). The cosmological results of the cosmology paper stand independently of the reduction attempted here.

Keywords: horizon thermodynamics; generalised entropy; Barrow entropy; holographic dark energy; de Sitter holography; entanglement entropy; non-equilibrium statistical mechanics; entropic gravity; dark energy

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1 Introduction

Horizon thermodynamics is the most robust pointer we have toward the microscopic structure of gravity. The first law on a causal horizon yields the Einstein equations [1], the apparent-horizon first law yields the Friedmann equations [2], and the Bekenstein–Hawking area law S = A/4 organises black-hole and cosmological thermodynamics alike. Yet the area law is an assumption about a *state*: it is the entanglement entropy of a particular (ground-like) horizon state, and generalisations — Tsallis–Cirto non-extensive entropy, Barrow's fractal-horizon $S \propto A^{1+\Delta/2}$ [3], and the holographic dark-energy programme built on them [4] — explore what happens when that assumption is relaxed.

Structural Entropy Dark Energy (SEDE; the cosmology paper [5]) is one such model, distinguished by sourcing dark energy from horizon entropy *gated by structure growth*,

$\rho_{\rm DE} = T_{\rm AH}\ s_{\rm grav}\ f_{\rm sat}(z)$. Following the cosmology paper, this is *holographic* dark energy — $\rho_{\rm DE}$ is added as a single fluid to an otherwise standard-GR Friedmann equation, not a Barrow-*modified*-gravity cosmology in which the horizon entropy reshapes the background; the volume-law entropy sources $\rho_{\rm DE}$ alone (so the modified-gravity Barrow/BBN bounds do not apply, and the postulate examined below is one of *state*, not of dynamics). Its appeal is that the entire dark sector adds no *fitted* parameter once $\Omega_{\rm m}$ and the one postulate are fixed: the H-coupling $\lambda = 1 - \Delta/2 = \frac{1}{2}$, the equation of state, the structure coupling γ, and the sound speed all follow. The one input is the **volume-law postulate**: the horizon entropy is the volume-law ($\Delta = 1$) entanglement entropy, not the area law. The cosmology paper states plainly that this is the model's single irreducible assumption, motivated but not derived, and that it is falsifiable — Δ is measured by upcoming surveys.

This paper asks the prior question: *how irreducible is it, really?* We are not trying to solve quantum gravity; we are trying to determine the minimal, sharply-posed statement that the cosmology actually rests on, and to reduce everything reducible to it. The result is a reformulation — one that ends with a shorter, sharper list of residuals than the postulate it replaces. We trade a static, untestable counting postulate for a physical picture in which $\Delta = 1$ is selected *at the horizon's birth*, carried through the ordering bifurcation that gravity's N-dependent coupling ($J/J_{\rm c} = 2\pi G m^2 N$) forces at Planckian size, with Barrow's entropic tilt breaking the tie and exact long-range hysteresis locking every horizon to its birth branch. The remaining hand-off to de Sitter holography is sharpened to a single algebraic question — the type, II_1 or II_∞, of the FRW static-patch observer algebra. Several steps borrow established machinery: the Cohen–Kaplan–Nelson holographic energy bound [6] for the scale; the Page curve and eigenstate thermalisation for the state; the Campa–Dauxois–Ruffo classification of long-range systems [7] for the connectivity; Ryu–Takayanagi min-cut for the count↔connectivity link; and the double-scaled Sachdev–Ye–Kitaev (SYK)/de Sitter correspondence [8, 9] for the closure attempt. The contribution is the chain, and the precise localisation of what remains open.

Throughout, every quantitative claim is a runnable experiment that the accompanying code runs with validation assertions.

1.1 Relation to prior work

SEDE sits at the confluence of several programmes, and its closest relative deserves explicit positioning.

Entropy-production cosmic acceleration (GREA). The General Relativistic Entropic Acceleration theory of García-Bellido and Espinosa-Portalés [10, 11] is the nearest existing model: it too abandons Λ, sources the late-time acceleration from cosmic-*horizon* entropy, takes the Helmholtz free energy $F = U - \mathrm{TS}$ (not matter alone) as the gravitational source, and makes the breaking of time-reversal by *entropy production* the engine of acceleration. We regard GREA as prior art for "acceleration from horizon entropy production without Λ", and our driven-NESS picture (§4–6) is conceptually continuous with it. The differences are sharp and testable: (i) GREA uses the boundary (area) horizon entropy and a single parameter α (the horizon-to-curvature ratio), whereas SEDE's defining claim is the volume entropy ($\Delta = 1$) — precisely the counting residue this paper isolates; (ii) GREA's entropy growth is driven by the horizon's own *expansion*, whereas SEDE's is *gated by structure formation* (the $f_{\rm sat}$ factor), which locks w(z) to the growth history and yields a phantom crossing rather than GREA's quintessence-like w(a); (iii) SEDE's dark sector adds no *fitted* parameter. Notably, were SEDE's residue to resolve to the *area* branch ($\Delta = 0$), its phenomenology would collapse toward GREA's — so the Δ measurement (§7) that decides SEDE's residue also discriminates the two models. A holographic reading of GREA has recently appeared [12], and a broader open-system view of gravitational non-

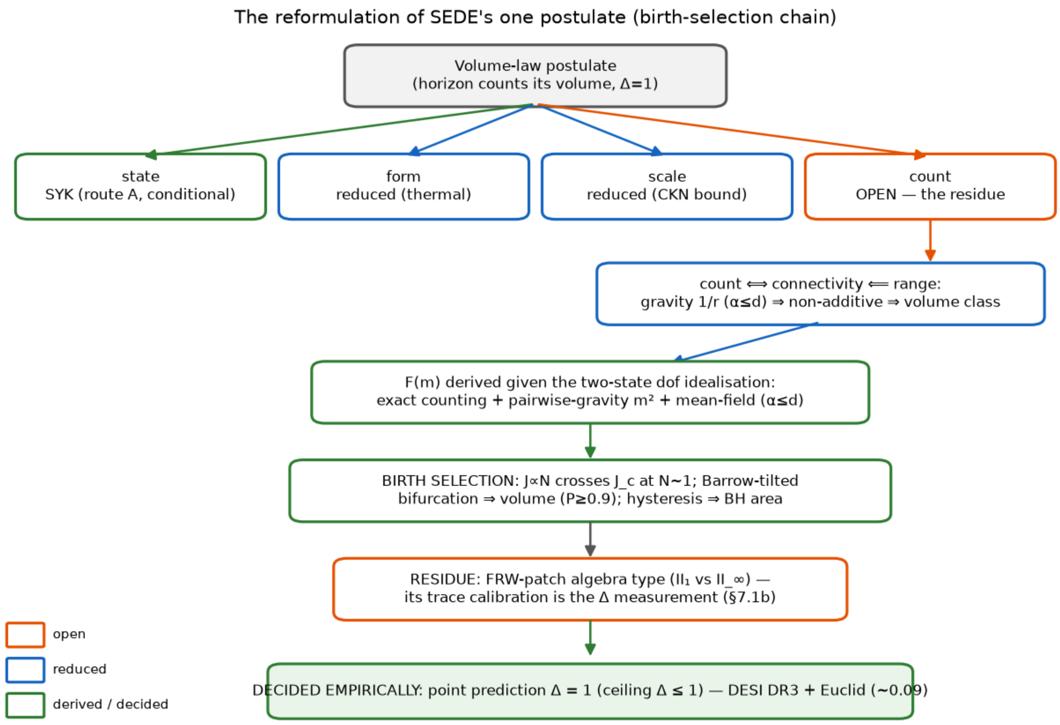

Figure 1: The reduction in outline. The volume-law postulate splits into *state*, *form*, *scale* and *count*; the first three are derived or reduced (green/blue) and the count is the residue (orange). The count reduces through connectivity and interaction range to a birth-branch selection: the N-dependent coupling carries the grown horizon through the ordering bifurcation onto the Barrow-tilted (volume) branch, where long-range hysteresis locks it (a *state-selection* hysteresis of the entropy branch — distinct from the unrelated "cosmological hysteresis" of cyclic cosmologies, which refers to the $\oint p\,\mathrm{d}V$ loop), leaving a single dS-holography question that the Δ measurement decides empirically. Each box names the experiment that establishes it.

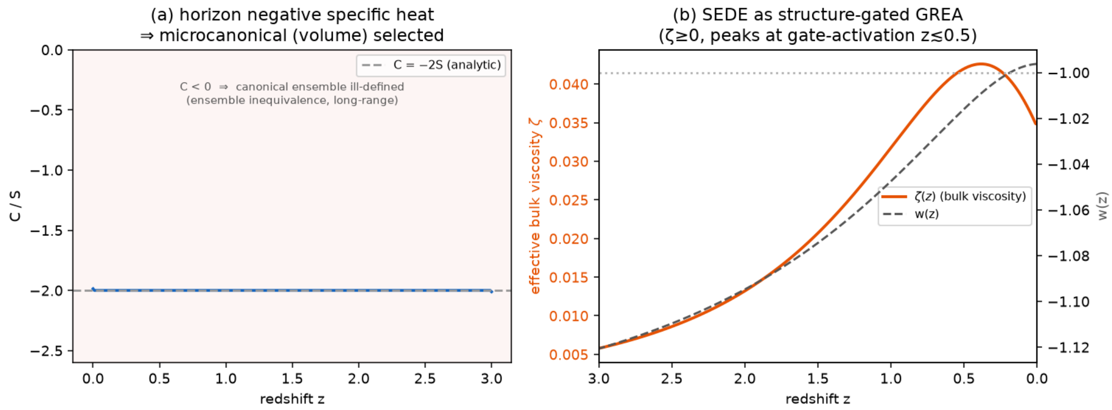

Figure 2: Two consequences drawn from the new literature. *(a)* The de Sitter horizon has negative specific heat ($C = -2S$); with gravity strongly long-range this makes the ensembles inequivalent and the canonical (area-law) accounting ill-defined, selecting the microcanonical (volume) branch (§4). *(b)* SEDE's structure-driven entropy production maps to a positive (second-law) effective bulk viscosity $\zeta(z)$ that peaks at the gate-activation epoch (z <∼ 0.5; it tracks d ln $f_{\rm sat}/d$ ln a, not the $z \approx 2$ star-formation peak), casting SEDE as a structure-gated member of the GREA/entropic-dark-energy family; w(z) crosses -1 at the viscous–dilution balance.

equilibrium thermodynamics is developing [13, 14]; SEDE's contribution to that line is the reduction of the volume-law *counting* input to a driven steady state. The connection is also quantitative: GREA's entropy production maps to an effective bulk viscosity with negative pressure, and SEDE's structure-driven entropy production d ln $f_{\rm sat}/d$ ln a > 0 maps to a *positive* (second-law-respecting) effective bulk viscosity $\zeta(z) = \rho_{\rm DE}$ (d ln $f_{\rm sat}/d$ ln a)/(9H) that peaks at the structure-formation epoch rather than tracking the horizon expansion; the $w = -1$ crossing is the GREA negative-pressure balance, structure-timed. SEDE is, in this sense, a *structure-gated* member of the GREA/entropic-dark-energy family — the gate being exactly what makes its w(z) lock to the growth history (the §5 phase-lock) rather than to the horizon's own size.

Holographic dark energy and generalised entropy. SEDE is, at the background level, a structure-gated Barrow holographic dark energy in the lineage founded by Li [15]: the CKN bound [6] fixes the scale, Barrow's fractal-horizon entropy [3] and its holographic-DE use [4] fix the form (the $\lambda = 1 - \Delta/2$ relation is Saridakis's; our increment is the $\Delta = 1$ selection, not that relation), and the Tsallis–Cirto non-extensive entropy [16] supplies the $\delta \leftrightarrow \Delta$ map. The CKN-vs-area-vs-volume tension these inputs create is mapped by Manoharan [17]; our addition is the *derivation* question — which of these inputs is irreducible — and the answer that only the count is.

Horizon thermodynamics and entanglement. The first-law derivations of Einstein [1] and Friedmann [2] equations underlie the construction; the entanglement-equilibrium refinement [18] is the basis of route B; the Ryu–Takayanagi min-cut [19] supplies the count $\Longleftrightarrow$ connectivity link; and the volume-law = thermal identification rests on eigenstate thermalisation [20–22].

de Sitter holography. The residue — "dim $\mathcal{H}$(static *patch*) $= e^{Area}$ or e^{Vol}?" — is posed in the language of de Sitter holography [23, 24] and is the target of the double-scaled-SYK/de Sitter correspondence [8, 9, 25], which we use in the closure attempt (§7). The maximal-scrambler state invokes the chaos bound [26] and SYK random-matrix universality [27].

Many-body and complexity inputs. The long-range/non-additivity classification is [7], and the α-controlled volume$\leftrightarrow$area (and intermediate/fractal) crossover of long-range entanglement entropy is already characterized in the literature [28–31] — the known crossover we apply to a horizon count (§ the count companion), not a new phenomenon. The prototype's free-fermion *negative* (volume-law needs interactions, not free transport) is consistent with the monitored free-fermion entanglement literature [32], and that genuine volume-law non-equilibrium steady states exist is established in driven many-body systems [33] — a condensed-matter precedent for, not a derivation of, the driven-NESS we invoke.

Structure-sourced dark energy. The general idea that dark energy is sourced by cosmic structure predates SEDE in several distinct mechanisms: Gough's information dark energy (≈ 2008); the spacetime-fragmentation/"FRW-islands" proposal of Hossain [34]; and the cosmological-backreaction and averaging programme (Buchert; Wiltshire's timescape). SEDE's cosmology paper credits and distinguishes these. What is specific to SEDE — and, to our knowledge, without precedent — is the *combination* the present paper analyses: the source is the volume-law horizon entropy ($\Delta = 1$, a counting claim), it is driven by structure through the f_{sat} gate, and the dark sector adds no *fitted* parameter. The closest entropic relative remains GREA (above), which differs in using the boundary (area) entropy. A systematic priority search returned no work combining these three elements; this paper concerns that horizon-entropy foundation. The most recent independent convergence is Pandey [35], who obtains late-time acceleration and suppressed growth from the *configuration entropy* of the collapsing matter field as a backreaction within GR — reaching SEDE's qualitative conclusion by a different route (matter-sector entropy, not the horizon; no derived w-crossing; no Δ), which we welcome as corroboration of the direction. A parallel generalised-mass-to-horizon-entropy programme [36, 37] links a modified horizon entropy to growth via a *modified-Friedmann* route, and so inherits the BBN Δ-bounds that SEDE's holographic-fluid scope evades.

2 The postulate, decomposed

"Volume-law" conflates four claims of very different status:

- state — the horizon degrees of freedom are maximally entangled (thermal), not in a low-entanglement ground state;
- form — the entropy scales as the volume, $S \propto V$;
- scale — the magnitude is $\rho_{\mathrm{DE}} \sim \rho_{\mathrm{crit}}$, not ρ_{Planck};
- count — the *number* of horizon dof grows as the bulk ($N \propto R^{d-1}$) rather than the boundary ($N \propto R^{d-2}$).

Since S $\sim$ (state factor) $\times$ N and the Barrow deformation enters as $S \propto A^{1+\Delta/2} \propto R^{(d-2)(1+\Delta/2)}$, the exponent Δ is fixed by the count (area $\to \Delta = 0$, volume $\to \Delta = 1$ in d=4); the state only sets whether the prefactor is maximal. So the postulate's empirical content — Δ — is the counting claim, and the other three are separable. The rest of the paper derives or reduces state, form and scale, and isolates count.

3 How derivable is the postulate? The route map

The horizon-entropy identification (assumed, not discharged). Routes A/B below characterise the *matter-field subsystem entanglement entropy* S_{ent} (Page/ETH/SYK) — UV-divergent, species-dependent. The dark sector is sourced by the *gravitational horizon entropy* $S_{\mathrm{grav}} = A/4G$ (Cai–Kim/Jacobson) — UV-finite, geometric. $S_{\mathrm{ent}} \equiv S_{\mathrm{grav}}$ is a long-standing open problem in quantum gravity ([38, 39]), established only for the Rindler vacuum. We adopt it (the induced-gravity premise) and flag every dependent step as conditional. In particular Route A fixes the entropy *prefactor* (maximal scrambling), which is orthogonal to the bulk-vs-boundary *exponent* that is the count; and Route B's "volume-law is generic" is a statement about S_{ent}, not S_{grav}. Accordingly the *state*, *form*, and residue-to-range steps are argued only conditionally on $S_{\mathrm{ent}} = S_{\mathrm{grav}}$ — not established outright.

Five routes (`run_deriv_{A..E}_*.py`):

A — de Sitter holography / maximal scrambler. A Majorana-SYK diagonalisation confirms the horizon *state* is maximally entangled / volume-law: chaotic level statistics ($\langle r\rangle \approx 0.71$) and Page-value saturation [40] ($S/S_{\mathrm{Page}} \approx 0.99$). But SYK is all-to-all / geometry-free and cannot pose the bulk-vs-boundary count; the count is the genuine open dS-holography sub-target.

B — entanglement first law in a thermal state. Jacobson's derivation gets the area-law Einstein equations from maximal entanglement of the *vacuum*. Redone around a *thermal* (de Sitter Gibbs) reference, the entanglement entropy acquires a volume term that dominates the area term once the region exceeds the thermal length $\lambda_{\mathrm{th}} = 1/T$. Volume-law is thus the generic thermalised behaviour, not exotic. At the de Sitter temperature alone the horizon sits just on the area side ($R_{\mathrm{H}}/\lambda_{\mathrm{th}} = 1/2\pi$); volume-law dominance requires thermalisation above the de Sitter bath (toward Planck = maximal scrambling), whereupon the density is Planckian — the CKN scale. So the *form* reduces to thermalisation, leaving the *scale* to CKN.

C — Verlinde emergent gravity. A volume-law de Sitter entanglement entropy is exactly what Verlinde's emergent-gravity programme invokes for apparent dark matter; one volume-law scale gives both $\rho_{\mathrm{DE}} \sim \rho_{\mathrm{crit}}$ and $a_0 = cH_0/2\pi$ ($\approx 0.87\times$ the radial-acceleration-relation value). A unification motivation that ties the postulate to an independent anomaly; it inherits that programme's open issues.

D — gravitational non-additivity. The Tsallis–Cirto map $\delta = 1 + \Delta/2$ makes $\Delta = 1 \iff \delta = 3/2$ exact, but pinning δ from a measurable non-additivity fails: real cluster kinematics are Gaussian ($q \approx 1.01$), not $q \approx 1.5$ — the weakest route.

E — no-go from energy bounds. The Bekenstein bound on the horizon energy gives $S \leq 2\pi \mathrm{RE}/\hbar\mathrm{c} = \frac{1}{4}(A/\ell_{\mathrm{P}}^2)$ — *exactly* the area law (verified $S_{\mathrm{Bek}}/S_{\mathrm{area}} = 1.000$, the known 10^{122}). The volume law therefore cannot come from any maximum-entropy/energy argument; it exceeds the bound by precisely $R/\ell_{\mathrm{P}} \sim 10^{61}$ (the "seam"). This proves the missing ingredient is a state/counting input and fixes its size.

Net: state (A) and form (B) are argued (conditional on $S_{\mathrm{ent}} = S_{\mathrm{grav}}$ — and "state" fixes the entropy *prefactor*, not the count *exponent*), the scale (CKN) is consistent with the data, the canonical area-law horn is genuinely removed by ensemble inequivalence (§4); the irreducible residue is the bulk-vs-boundary count.

4 Reducing the residue to a driven NESS

The count is not a free coin-flip — but here we must be careful to avoid a circularity, because the entanglement class cannot simply be *posited.* It is fixed by the Hamiltonian and the state, and the decisive fact is that the entanglement area law is a theorem with a hypothesis: it is proved for *local* (finite-range) Hamiltonians — Hastings [41] rigorously in 1D via Lieb–Robinson bounds, Brandão–Horodecki [42] from exponential decay of correlations, reviewed in Eisert–Cramer–Plenio [43] — and the proof *requires* exponential clustering. Gravity is 1/r ($\alpha = 1$), strongly long-range ($\alpha \leq d$) and non-additive [7], and there the hypothesis fails: in an explicit gapped free-fermion model we find the correlation function decays as a *power law* for $\alpha = 1$ (slope -1.5) versus *exponentially* for the short-range reference (*slope* $-\, 6.9$), exactly as Koffel–Lewenstein– Tagliacozzo [44] found. So the area-law theorem does not apply to a gravitationally-bound horizon: area-law is *not guaranteed*, and the count is genuinely open between area and volume with neither the default. We are explicit about what this does and does not give. It does *not* by itself derive volume — a noncritical long-range ground state can still be area-law [45], and indeed the gapped ground-state block entropy stays bounded for every α we test. The volume law additionally requires the horizon *state* to be thermal / maximally scrambled (route A), where entanglement is volume-law; and even that fixes the *state*, not the *count.* The super-extensive non-additivity ($u \propto N^{1-\alpha/d}$, slopes 0.53 in d=2, 0.68 in d=3 vs 0.50, 0.67) is the thermodynamic face of the same long-range property. The conclusion is that volume is the *motivated, non-default* class for a self-gravitating thermal horizon — area-law arises only when the holographic bound intervenes for an *isolated, equilibrium* horizon (a black hole saturates the Bekenstein bound, route E, and relaxes to area-law) — while the bulk-vs-boundary count itself remains the open dS-holography residue. One caveat qualifies this: the area-law theorems are statements about many-body systems with a specified Hilbert space and Hamiltonian, and what the horizon's degrees of freedom *are* — and what Hamiltonian governs them — is precisely the open dS-holography question. So this is a motivated *conditional* — *if* the horizon is a long-range many-body system, *then* area-law is not forced and volume is favoured — not a theorem about the actual horizon. What it establishes is that the entanglement class is read off the (modelled) Hamiltonian rather than assumed — which is what defeats the circularity charge — not that the count is derived.

This same long-range property has a second, independent consequence that bears on the canonical-vs-microcanonical λ ambiguity (whether $\rho_{\rm DE} \propto H^2\, f_{\rm sat}$, area, or $\propto H\, f_{\rm sat}$, volume). A non-additive system has *inequivalent* statistical ensembles [7], the signature being negative specific heat — and the de Sitter horizon indeed has $C = -2S < 0$ throughout its history. For such a system the canonical ensemble is ill-defined; the isolated self-gravitating horizon must be treated *microcanonically* (by its state count). The canonical accounting is precisely the area-law ($\lambda = 1$) branch, so ensemble inequivalence removes it as unphysical and selects the microcanonical (volume, $\lambda = 1/2$) branch the model uses. This does not close the residue — whether the microcanonical state count is itself volume is the dS-holography question — but it eliminates one of its two horns from first principles. The cosmic horizon differs in one respect: it is continuously *driven* by structure formation (the $f_{\rm sat}$ gate, $dS_{\rm struct}/dt > 0$; present drive ≈ 0.45). This turns the black-hole/cosmic asymmetry into a discriminator — equilibrium $\to$ area, driven NESS $\to$ volume — and reduces the residue to a single, falsifiable statement:

> **Driven-NESS conjecture.** A strongly-long-range horizon, driven by structure formation, settles in the volume-law steady state rather than relaxing to the area-law equilibrium.

Two sections sharpen this conjecture into calculations. The *selection* half is replaced by a mechanism that needs no steady-state argument at all — birth at the ordering bifurcation, §5 (iii) — while the *driving* half is localised in a one-line activation kinetics whose solution is exactly the cosmology paper's gate (§7.1) [5]: the conjecture survives as the statement that structure deposits *populate* the volume capacity, not that they select the branch.

5 Selection of the volume branch: birth at the ordering bifurcation

We ask what selects the volume branch dynamically, separating what is robust from what is conditional. Four results, each backed by an explicit calculation.

(i) Barrow entropy supplies a driving slope, not a landscape. The bare Barrow free energy $F(\Delta) = E - T_{\mathrm{AH}}\, S_\Delta$, with $S_\Delta = (A/A_0)^{1+\Delta/2}$ and the Misner–Sharp energy, is *monotonic* in Δ — F(0) = 0 by the first law, and F decreases to $\Delta = 1$ with no barrier and no second minimum. Bistability is therefore not a consequence of Barrow entropy: Barrow supplies the entropic drive toward $\Delta = 1$, not the two-phase structure. (This is the actual content of the GSL argument of §8.1 — it drives $\Delta \to 1$, it does not by itself produce a spinodal.)

(ii) Gravity supplies the barrier — deep in the ordered phase. A two-phase (area$\leftrightarrow$volume) free energy is bistable only if the horizon degrees of freedom have a cooperative coupling above the ordering threshold, $J \geq J_{\mathrm{c}}$. That coupling is the largest eigenvalue of the gravitational coupling matrix, $J = \lambda_{\max}(W_{\mathrm{grav}})$, $W_{\mathrm{ij}} = g/r_{\mathrm{ij}}^{\alpha}$ — the *same* super-extensive quantity behind the non-additivity of §4 ($\lambda_{\max}/row$-sum = 1.02), so volume-counting and volume-locking are one number. Comparing it to $J_{\mathrm{c}} = T_{\mathrm{AH}}$ *in the same units*, for N $\sim$ A/4 Planckian dof on the horizon 2-sphere, gives $J/J_{\mathrm{c}} = 2\pi G m^2 N \sim 2\pi N \sim 10^{123}$ (the non-additive scaling $\lambda_{\max} \propto N^{1-\alpha/d} = N^{1/2}$ is confirmed numerically — slopes 0.52/0.67, a short-range $\alpha > d$ control bounded at $N^{0.01}$). Gravity clears the threshold by of order the holographic count: the two-phase well exists, its minima are the pure-area and pure-volume phases, and the barrier is $\sim N \cdot T_{\mathrm{AH}}$. This places the horizon deep in the ordered phase, not near a spinodal — *today.* But the same formula contains the key to the selection: $J/J_{\mathrm{c}} = 2\pi G m^2 N$ is N-dependent, so the barrier did not exist at the horizon's birth. The Landau form is not otherwise assumed *given the two-state dof idealisation* — the one place the $S_{\mathrm{ent}} = S_{\mathrm{grav}}$ tier re-enters, sourced micro-structurally below, and the qualification that governs everything in this item: within that idealisation all three terms of $F(m)/\mathrm{NT} = -(J/2)m^2 + \theta m +$ [m ln $m + (1-m)ln(1-m)$] are derived — the mixing entropy is exact combinatorics of choosing which dof participate; the m^2 term is the pairwise 1/r binding of the participating dof (measured exponent 2.008), with $J = \lambda_{\max}(W_{\mathrm{grav}})$ the same number as above; and mean-field thermodynamics is *exact* for $\alpha < d$ [7] (verified: a Kac-normalised $\alpha = 1$ long-range Ising sits at the mean-field T_{c} while the nearest-neighbour control sits at Onsager's, far below). Even the two-state (boundary-/bulk-entangling) character of the dof is not an idealisation inserted by hand: in the random-tensor-network model of holographic states [46] the entanglement computation is *exactly* dual to an Ising model — each tensor carries a binary (identity/swap) replica variable whose domain wall is the Ryu–Takayanagi min-cut — and we verify both the min-cut law (a three-leg region behind a bulk

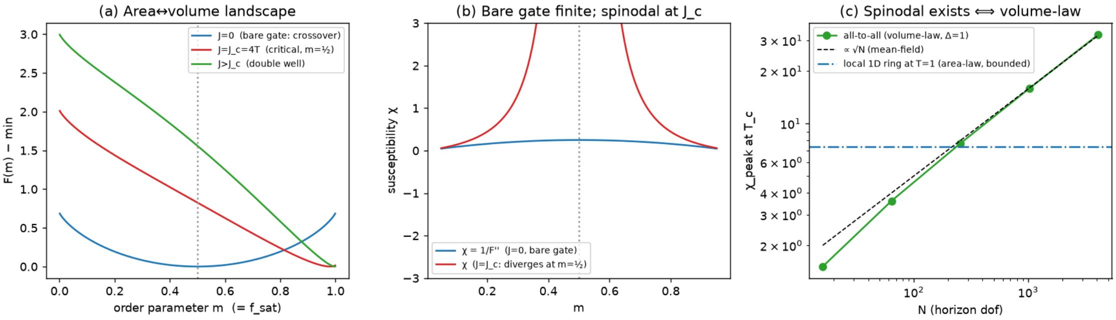

Figure 3: The two-phase horizon free energy. *(a)* Smooth crossover for the bare gate (J = 0); a spinodal develops only for $J \geq J_c$. *(b)* The bare-gate susceptibility is finite; it diverges at the spinodal. *(c)* The cooperative coupling $J = \lambda_{\max}(W_{\mathrm{grav}})$ is the gravitational binding itself: all-to-all (volume-law) connectivity gives $\chi_{\mathrm{peak}} \propto \sqrt{N}$ (so $J/J_c \sim$ N, far above threshold), while local (area-law) connectivity is bounded. Gravity thus makes the well bistable — but *deeply* so today, which is why selection happens at the birth bifurcation (result iii, Fig. 4) rather than by barrier crossing (the horizon's membrane hydrodynamics, result iv, is a separate matter).

bottleneck saturates at $1 \cdot ln\ \chi$, not $3 \cdot ln\ \chi$) and the duality itself (the two-state partition sum Z_A/Z_0 reproduces the measured purity to $< 1\%$ at bond dimension $\chi \geq 4$). The order parameter thereby acquires a microscopic definition — m = the fraction of horizon bonds on the bulk side of the min-cut — and the residual assumption shrinks to the tensor-network description of the horizon state itself: the $S_{\mathrm{ent}} = S_{\mathrm{grav}}$ tier restated micro-structurally, one assumption where two stood before.

(iii) Selection at the ordering bifurcation, and exact hysteresis. A barrier $\sim N \cdot T_{\mathrm{AH}}$ is *uncrossable*: Kramers escape is suppressed by $e^{-N \cdot \Delta f}$, N $\sim 10^{122}$ — so "relaxation to the global minimum" cannot be the selection mechanism, and we do not invoke it. Instead, the N-dependence of (ii) supplies the mechanism. The cosmic apparent horizon grew continuously from N $\sim$ O(1), so it passed *through* the ordering bifurcation as the double well formed around it: near threshold the barrier vanishes quadratically, $\Delta f \propto (J - J_c)^2$ (measured exponent 1.89), while the Barrow tilt of (i) is already present — and a *tilted* pitchfork has no branch ambiguity. The minimum continuously connected to the disordered N $\sim$ 1 state is the tilt-favoured (volume) branch; the area minimum is born *metastable* only at the spinodal $J_{\mathrm{sp}} > J_c$ ($J_{\mathrm{sp}} \approx 5.6$ at tilt b = 0.3) and is never occupied. The swept trajectory confirms it: gradient flow tracks continuously into the volume well (final m = 1.00, no discontinuity) — $\Delta = 1$ is selected at birth, with no barrier ever crossed. The corollary is as important as the result: for long-range/mean-field systems metastability is *exact* — there is no finite nucleation droplet [47], so every horizon keeps its birth branch forever. A black hole, created abruptly at macroscopic N by collapse of a low-entanglement (vacuum-like) region, is born *in the area well* and locked there ($N \cdot \Delta f \sim 10^{77}$ for a stellar hole, against $ln(t_0/t_P) \approx 140$): $\Delta_{\mathrm{BH}} = 0$ is derived, and the same e^{-N} suppression that forbids relaxation-based selection protects the black-hole prediction of §7.1. The 10^{-7}-drive vs 10^{61}-dof objection never arises, because nothing is ever flipped. *Flag, quantified:* the mean-field landscape is here extrapolated to N $\sim$ O(1)

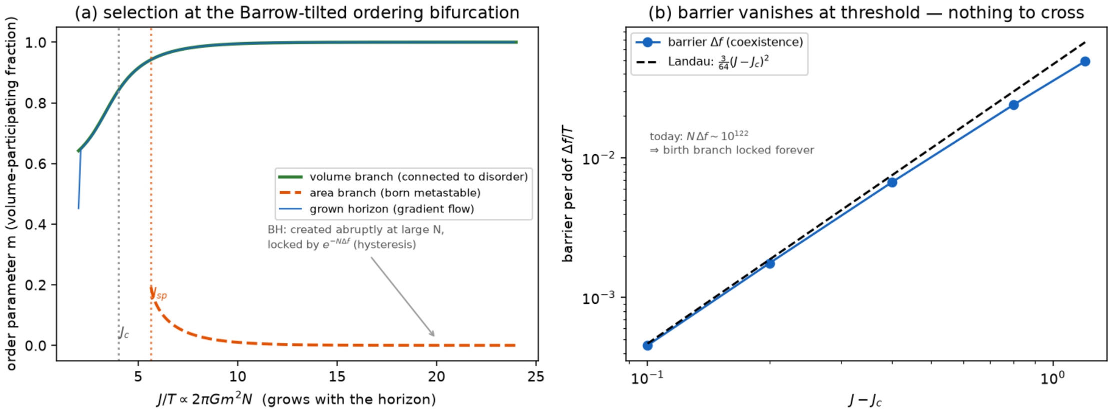

Figure 4: Selection at the ordering bifurcation. *(a)* With the Barrow tilt, the minimum continuously connected to the disordered state is the volume branch (green); the area branch (orange, dashed) is born *metastable* only at the spinodal $J_{sp} > J_c$ and is never occupied by the grown horizon, whose gradient-flow trajectory (blue) tracks the volume branch with no discontinuity. A black hole, created abruptly at large N, starts on the area branch and is locked there by exact long-range hysteresis. *(b)* The barrier vanishes quadratically at threshold — Landau's $3(J-J_c)^2/64$ — so the grown horizon crossed the bifurcation when there was nothing to cross; today's $N \cdot \Delta f \sim 10^{122}$ locks the birth branch forever.

— the quantum-gravity era — and we stress-test rather than assume the adiabaticity. A Kibble–Zurek scan of the swept, tilted bifurcation with per-dof noise $\sim 1/N$ finds: (a) at the *physical* Barrow tilt $b_{phys} = \sqrt{N} - 1$, selection is deterministic (P = 1.000) for $N \geq 10$ at every sweep rate; (b) the untilted control is a fair coin, confirming Barrow's slope (i) as the selecting agent; (c) in the fully self-consistent sweep ($J = 2\pi N$, so the horizon is born already weakly ordered), selection is equilibration-limited — P(volume) rises monotonically from 0.64 (absurdly fast) toward 1 with sweep time, reaching ≈ 0.9 at the slowest rates tested, and is volume-biased in *every* regime. The physical clock ($\geq N^{3/2}$ relaxation times per e-fold of growth) sits on the slow/equilibrated side, but its O(1) normalisation at N $\sim$ 2 is genuinely Planck-era physics: the precise form of the selection claim is therefore a *probability* ≥ 0.9 rising to 1 in the equilibration limit, never a coin flip, and never area-biased — with the Δ measurement (§7) testing the outcome for the one horizon we have.

(iv) The horizon's membrane hydrodynamics — and why no *observable* z* transient is claimed. The apparent horizon has first-principles hydrodynamics: transverse redistribution is locally conserving (the Damour–Navier–Stokes membrane equation, with the *dynamical/apparent*-horizon bulk viscosity positive, $\zeta = +1/16\pi G$ [48], unlike the teleological event horizon's negative value), and two coefficients are *exact* — the sink rate, derived symbolically from the Cai–Kim definition $\kappa = (1/2\sqrt{-h})\partial_a(\sqrt{-h}\, h^{ab}\partial_b\, \tilde{R})$ on the flat-FRW apparent horizon, $\kappa = -H(1-3w_{eff})/4$ (H/4 in the matter era, H in de Sitter, zero at radiation domination), and the deposition source, from the perturbed Misner–Sharp condition, $\delta R_A/R_A{=}{-}\frac{1}{2}{\cdot}\delta M/M$. One might read these coefficients as the dissipation of a *roughening* horizon surface and infer a near-conservative, self-organised-critical z* transient — a truncated $\tau \approx 3/2$ avalanche window and a deterministic ϵ-lag in $f\sigma 8$. **We do not claim such an observable** (the case is set out in §6). The companion count paper [49] shows the classical apparent-horizon *shape* is constraint-slaved: the $\theta_+ = 0$ condition is elliptic ($\nabla^2_S h + 2h + 2h^2 = 2\mathcal{S}$, its eq. 2.9), with no $(\nabla h)^2$ and no ∂_t h at any order, so the horizon has no growth equation and does not roughen — there is no height field to carry an avalanche dynamics, and the count paper's bond-cut analysis finds no roughening universality class either. The membrane coefficients above stand as exact GR facts, but they do *not* source a critical z* observable. What survives is the *drive* itself (next paragraph), which populates the volume capacity selected at birth; the near-critical/SOC layer and the ϵ-lag it would carry are not claimed (§6).

The drive is a violent-relaxation deposit, and the γ-mechanism is the same event . Structure forms by gravitational collapse, i.e. by *violent relaxation* [50], which drives a long-range system on the dynamical time to a quasi-stationary state rather than to thermal equilibrium. This single event does two things SEDE otherwise treats separately. First, it dissipates the halo binding energy $E_{\rm bind} = M\sigma_{\rm v}^2 \propto M^{5/3}$ (virial: $R_{\rm vir} \propto M^{1/3}$, $\sigma_{\rm v}^2 \propto M^{2/3}$) into the horizon ledger — the mass weight that *is* the structure coupling $\gamma \approx 1.5$ (the cosmology paper, Prescription 4C) [5]. Second, that same deposited binding energy ($\propto \langle\sigma_{\rm v}^2\rangle$) is the amplitude of the structure drive that populates the volume capacity. So the γ-weight and the drive are one quantity from one process — this identification is robust, and independent of the excluded near-critical layer (result iv). Two corollaries follow: collapse is prompt ($t_{\rm dyn}/t_{\rm Hubble} = \Delta_{\rm vir}^{-1/2} \approx 0.07$), so the horizon is driven by an ongoing sequence of such events whose envelope is $f_{\rm sat}$; and, once the volume branch is selected, the quasi-stationary state's lifetime in a long-range system scales as N^δ [7], so at the horizon's N $\sim 10^{122}$ it is permanent — a second permanence argument beside the second-law one (route E). Both the halo QSS and the de Sitter horizon have negative specific heat, consistent with the microcanonical (volume) selection of §4.

6 The z* transient: why none is claimed

It is natural to ask whether the structure drive produces a *distinctive* observable at z. *If the drive arrived on a nearly conservative* roughening* horizon (§5 iv), the z* response would carry mean-field self-organised-criticality statistics — a scale-free $\tau \approx 3/2$ avalanche window, truncated at a computable cutoff $s_c = 2/\epsilon^2$ — whose survey-accessible face would be a deterministic ϵ-lag in $f\sigma 8$ (a z*-localised, growth-locked distortion, $S/N \approx 0.5$–1.5). **We claim no such observable**, for a first-principles reason. The companion count paper [49] shows the classical apparent horizon is constraint-slaved and has no growth equation (its eq. 2.9: $\theta_+ = 0$ is elliptic, with no $(\nabla h)^2$ and no ∂_t h at any order), so it does not roughen; there is no height field, no roughening universality class, and hence no self-organised-critical avalanche dynamics of the horizon and no ϵ-lag derived from one.

The exclusion is surgical. It does **not** touch the selection of $\Delta = 1$, which is fixed at the horizon's birth (§5 iii) independently of any z* dynamics, nor the exact membrane coefficients of §5 (iv) ($\kappa = -H(1 - 3w_{\rm eff})/4$, $\delta R_A/R_A = -\delta M/2M$), which stand as GR facts. The empirical cost is minimal: such an ϵ-lag would be marginal ($S/N \approx 0.5$–1.5) and the scale-free layer is in any case *not* two-point observable, so the model loses no decisive test — the decisive prediction remains the point value $\Delta = 1$ (§7), decided by DESI DR3 + Euclid. The frozen falsifier matrix carries no z* two-point entry accordingly. The parent predictions carried by the cosmology paper (§6) [5] (the CHR enhancement, P3–P5) rest on the same roughening premise and receive the same treatment; that reconciliation is separate and is not taken here.

7 What would make it a closure

A closure derives F(m) itself, reducing the dark sector to Ω_m + *established* physics. The three routes converge:

- **A (DSSYK).** Entropy extensive in the dof count ($S_{max} = (N/2)ln\ 2 \propto N$, slope 1.00); cannot be super-extensive ($volume \propto N^{3/2}$). We state this plainly: read in the standard Narovlansky–Verlinde dictionary, the leading concrete model of de Sitter holography maps to the de Sitter *area* (Gibbons–Hawking) — i.e. it currently favours the area branch, a real tension SEDE's volume postulate must eventually overcome. The mitigating point — that DSSYK is all-to-all / geometry-free and so cannot, by itself, *pose* the bulk-vs-boundary counting question — is secondary, not a neutraliser. As things stand, route A is evidence to be answered, not used.
- **B (Euclidean dS saddles).** Einstein and Barrow horizon free energies each have a single saddle — no volume-law saddle of the bare path integral (route-E no-go in saddle form).
- **C (thermal F(m)).** The bistable landscape *is* derivable — $F(m)/T = -(J/2)m^2 + \theta m + [m\ \ln\ m + (1-m)ln(1-m)]$ from gravitational cooperativity ($J = \lambda_{max} \geq J_c$), the holographic pull (route E), and configurational entropy: area well $m \approx 0.07$, volume well $m \approx 0.93$ whenever $J \geq J_c$ — and each term is now sourced independently (exact counting; pairwise-gravity m^2; mean-field exactness for $\alpha < d$), so route C's status is upgraded from "assumed form" to "derived given the two-state dof idealisation". Level-2 existence, conditional on the volume branch being accessible.

All three reduce to one statement — *the horizon's dof count is the bulk (volume) one* — which A and B cannot supply and C uses. Closure has three precise forms: **Level 1**, a dS-holography computation of the bulk count (open, not shortcut-able past route E; posed algebraically in §7.1b); **Level 2**, exhibiting the volume branch as a real saddle/state (C gives the landscape conditional on it); **Level 3**, *measuring* Δ (DESI DR3 + Euclid to $\sim$0.09; the pre-registered point $\Delta = 1$ establishes, $\Delta = 0$ refutes, an intermediate value falsifies the locked-Δ picture — decisive, in hand; the predictions are frozen with an amendment protocol in the pre-registration).

Level 3 already has a present-data instalment. On DESI DR2 BAO + Pantheon+ + CMB distance priors + cosmic chronometers $+f\sigma 8$, in one CAMB-in-the-loop pipeline, the un-gated Hubble-cutoff Barrow family pays $\Delta\chi^2 = +2416$ to sit at $\Delta = 1$ — reproducing, in this related family, the community result that bare Barrow HDE cannot live at the volume endpoint [51]. The gated family, by contrast, pays only $\Delta\chi^2 = +0.8$, with a profile-likelihood interval $\Delta = 0.93$, 68% CL [0.83, 1.02] (all base parameters re-optimised at each Δ — a profile, *not* the full marginalised posterior, which remains the DR3 + Euclid target; the label is precise because the cosmology paper (§5.6) [5] reserves "marginalised measurement" for that future). The pre-registered window lies inside the interval, and $\Delta = 0$ is excluded by $\Delta\chi^2 \approx 371$ — a *profile* statement: read as such, it overstates the marginalised model-level signal (the cosmology paper's ΔDIC ≈ -3) by roughly two orders of magnitude in χ^2, and must not be quoted as a posterior exclusion. SEDE at $\Delta = 1$, carrying zero dark-sector parameters, fits these data 2.9 χ^2-units *better* than ΛCDM at equal parameter count, and within 1.4 AIC-units of w0waCDM, which spends two extra parameters. The structure gate is, quantitatively, what makes the volume-law count observationally viable — on data already public.

7.1 The count is state-dependent — and that resolves the "everything points to area" tension

Routes A and B, and the standard handles generally (the de Sitter modular Hamiltonian is a *local* boost; DSSYK in the Narovlansky–Verlinde dictionary; the CKN holographic bound; black-hole thermodynamics), all favour the area count. Read as universal statements about "the" horizon count they look like a standing objection. The resolution is that they are not universal: the count is state-dependent, and every one of those handles is an *equilibrium* horizon. The equilibrium-vs-driven discriminator of §4 is exactly this — the f_{sat} gate *is* the fraction of the volume capacity that structure-driving activates, $N_{\text{eff}} = N_{\text{area}} + (N_{\text{vol}} - N_{\text{area}}) \cdot f_{\text{sat}}$, so the equilibrium limit ($f_{\text{sat}} \to 0$) is area and the driven, locked limit ($f_{\text{sat}} \to 1$) is volume . (This is *not* the rejected running-Δ background of §3: the cosmic horizon is carried through the ordering bifurcation at birth and *locks* at constant $\Delta = 1$ (§5 iii); the state-dependence is *across horizons*, driven cosmic vs equilibrium black hole, not a running Δ over cosmic time.) So the area results are SEDE's equilibrium baseline, not counter-evidence. One case needs care because it is *not* obviously equilibrium: the DSSYK correspondence is a duality with the de Sitter static patch — the cosmological horizon itself. The point is that it is a duality with *eternal* (matter-free) de Sitter, an equilibrium spacetime with no structure to drive it ($f_{\text{sat}} \equiv 0$); the real universe's apparent horizon is embedded in a matter-plus-structure cosmology and is driven. So "DSSYK = area" is the area of *equilibrium* de Sitter — again the f_{sat} $\to 0$ limit — and is consistent with a volume-law *driven* horizon, not a contradiction of it. Two derivations remove what used to be this argument's "one layer out" hedge. (a) The gate is kinetics, not a convention. The minimal absorption law — each violent-relaxation deposit activates a fraction of the *remaining* inactive volume capacity in proportion to the deposit, df = $\gamma(1-f)\ d(D^2)$ with f(D→0) = 0 — has the solution $f = 1 - e^{-\gamma D^2}$: *exactly* the cosmology paper's gate, verified against the real growth history to 10^{-5}. The interpolation $N_{\text{eff}} = N_{\text{area}} + (N_{\text{vol}} - N_{\text{area}}) \cdot f_{\text{sat}}$ is then the *occupancy* statement of the same kinetics, not an independent assumption, and the equilibrium horizons (black hole, eternal dS: $D \equiv 0$) are its f = 0 limit with no extension required. The alternative reading — f_{sat} as the *equilibrium* global-minimum map m(θ) *of the §5 landscape — is rejected by shape: deep in the ordered phase that map switches discontinuously at coexistence (a step, max slope ∼300 vs the gate's ∼2), so the smooth, growth-tracking gate can only be a* cumulative kinetic* quantity — which is the driven-NESS picture stated mechanically. Under the birth-selection of §5 (iii) the division of labour is clean: the *branch* is fixed at birth; f_{sat} is the *occupancy* of the volume capacity, not the branch selector. (b) The equilibrium-area ceiling is an algebra type. In the crossed-product analysis of Chandrasekaran, Longo, Penington & Witten [52], the *eternal* dS static patch (with observer) has a Type II_1 algebra — the unique case possessing a maximum-entropy state, and that maximum *is* the Gibbons–Hawking area — while the black-hole exterior is already Type II_∞ (entropy unbounded above). "$S \leq Area$" is thus the II_1 ceiling of the *equilibrium* algebra, not a law of horizons. Two further results sharpen this to its final form. First, a negative result: the algebra *type* is fixed by the infrared attractor — bounded horizon entropy (any accelerating attractor, ΛCDM *and* SEDE alike, since SEDE too ends at finite $E_\infty = \Omega_{\text{DE}}$) is II_1-compatible, while only eternally decelerating cosmologies force II_∞ — so the type does not discriminate the counts. Second, and decisively: in a Type II factor the trace — and hence the *value* of the maximal entropy — carries a free normalisation. CLPW fix it by calibrating to the Gibbons–Hawking entropy of the semiclassical Einstein saddle (their entropies are explicitly defined up to an additive constant). "The eternal-patch ceiling is the area" is therefore an Einstein-saddle calibration, not an algebraic output — the same II_1 structure calibrated to the volume count is equally consistent, matched to a modified gravitational sector. The residue thereby stops being an independent open question: the algebraic unknown is the trace calibration, the calibration is the count, and the count is Δ — the Level-3 measurement answers the Level-1 question restated (a standalone treatment is given in the companion trace-calibration note). The DSSYK corollary sharpens likewise: dim $\mathcal{H} = 2^{N/2}$ *by construction*, so DSSYK cannot pose super-extensivity for any state; whether its N scales as R^2 or R^3 is a UV input to the dictionary, not an SYK output.

This makes the residue doubly falsifiable, on two independent horizons. (i) The cosmic horizon (driven) should have $\Delta = 1$ — DESI DR3 + Euclid. (ii) The black hole (equilibrium) should have $\Delta = 0$ — and the observable is primordial-black-hole evaporation: SEDE *predicts that PBHs evaporate normally* (standard Hawking), because black holes are undriven, whereas a *universal*-volume theory forbids it ($T_{\rm B}/T_{\rm H} \sim 7 \times 10^{-21}$, evaporation time $\times\ 10^{81}$). The PBH channel therefore tests the *mechanism* — it distinguishes driven-NESS SEDE ($\Delta_{\rm BH} = 0$) from universal-volume quantum gravity ($\Delta_{\rm BH} = 1$) — and the black hole being area-law, far from a problem, is a SEDE prediction. This is a *future* discriminator, not present-day evidence: SEDE's $\Delta_{\rm BH} = 0$ means standard Hawking evaporation, consistent with all current data (including $\sim 10^{15}$ g PBH abundance bounds) precisely because it predicts no new black-hole effect; the channel discriminates only on a positive PBH-evaporation detection or a robust population statement.

This also yields a new statement about black holes in general: area-law is the *equilibrium* default, not a fundamental geometric law. A black hole is area-law *because of how it is born*: created abruptly at macroscopic N from a low-entanglement (vacuum-like) region, it starts in the area well, and exact long-range hysteresis (§5 iii) locks it there with Kramers exponent $N \cdot \Delta f \sim 10^{77}$. It is also undriven ($f_{\rm sat} \to 0$), so neither mechanism ever populates its volume capacity. The cosmic horizon's uniqueness is a calculation, not an assumption: it is the only horizon *grown* continuously from Planckian size through the ordering bifurcation. The most-driven black hole conceivable is a primordial one *at formation*, where the entire horizon is built by a collapsing overdensity (violent relaxation); yet even there the structure entropy deposited is negligible against the black-hole entropy *created*, $S_{\rm rad}/S_{\rm BH} \sim (M/M_{\rm P})^{-1/2}$, so the formation drive $f_{\rm form} \sim (M_{\rm P}/M)^{1/2} \ll h_{\rm c}$, the gate-activation threshold, by many orders. Accreting and merging black holes are weaker still. So SEDE robustly predicts *all* black holes are area-law ($\Delta_{\rm BH} = 0$), and the cosmic apparent horizon — uniquely driven *sustainedly* to $f_{\rm sat} \to 1$ over the whole structure-formation history — is the only volume-law horizon. The falsifier is correspondingly sharp: an *isolated* volume-law black hole (anomalously slow PBH evaporation, or a modified-area GW ringdown) would mean Δ is *universal* (Barrow-type) and would refute the state-dependent, driven-vs-equilibrium mechanism — so SEDE is *more* falsifiable on black holes than a universal-Barrow theory, not less.

The theoretical residue is correspondingly narrowed: not "derive the count," but two named statements — "the branch is set at the birth bifurcation" (§5 iii, derived within the F(m) landscape, N $\sim$ 1 extrapolation flagged) and "the deposits populate the volume capacity" (the gate kinetics, §7.1a) — both now *consistent with* the equilibrium area results (their $f_{\rm sat} \to 0$ / abrupt-birth limit), with the two-channel measurement deciding it either way regardless of dS holography.

Two final remarks, stated openly. First, the *magnitude* worry — a 10^{-7} structure drive against a 10^{61} change in the activated count — is dissolved rather than answered: under §5 (iii) the drive never flips anything. The branch (the 10^{61} capacity) is fixed at birth by the bifurcation; the 10^{-7}-per-Hubble- time deposits merely *populate* it, cumulatively, through the derived gate kinetics (§7.1a), and the two numbers no longer need to be compared. What remains modelled is the landscape's two-state idealisation (§5 ii), not a heavy-lifting amplification. Second, the *pre-registered* case: the prediction is a *point*, $\Delta = 1$, fixed by the birth-selection of §5 (iii) — not a window read off a roughening exponent. A measured cosmic Δ significantly below it — say $\Delta \approx 0.5$ from DESI DR3 + Euclid, $\sim 5\sigma$ from $\Delta = 1$ at $\sigma_\Delta \approx 0.09$ — would falsify the locked-$\Delta = 1$ picture outright and would *not* be a free parameter to absorb.

The count as a horizon Hausdorff dimension — and why the roughening reading is superseded. The Level-1 question — is dim $\mathcal{H}$(de Sitter static *patch*) $= e^{Area/4}$ or e^{Volume}? — has a clean geometric form: dim $\mathcal{H} = e^{Area} \iff$ the horizon is a smooth 2-surface (Hausdorff dimension $d_\mathrm{H} = 2$), and dim $\mathcal{H} = e^{Volume} \iff$ a space-filling fractal ($d_\mathrm{H} = 3$), so the count *is* the horizon's Hausdorff dimension, $\Delta = d_\mathrm{H} - 2$ — a reframing of "area or volume?" as an empirically-decidable *geometric* statement. One could try to turn that into a *dynamical* ceiling by treating the driven horizon as a roughening surface (a roughening 2-surface has $d_\mathrm{H} \in [2, 3]$, so $\Delta \leq 1$). That roughening reading is **superseded** by the companion count paper [49], which does the same job more strongly and without the surface-height picture that §6 sets aside: (i) the ceiling $\Delta \leq 1$ is a *theorem* from a finite per-site leg budget (bond-cut capacity $C \leq \kappa{\cdot}|\Omega|$), not a fractal argument — matching the bound the cosmology paper (§8.1) [5] claims and the dof-*counting* reading of §8.5; (ii) the count is *discrete*, $\Delta \in 0, 1$ — the count paper finds no minimum-cut surface for a long-range network, hence no height field and no roughening universality class, so Δ **is a count, not a Hausdorff dimension** (exactly the cosmology paper's §8.5, and confirmed classically by the constraint-slaved apparent horizon, eq. 2.9); and (iii) $\Delta = 1$ is the infinite-cutoff ($\mathrm{L} \to \infty$) endpoint for the flat, infinite spatial slice — a stated cutoff choice in that paper, not a derivation of the value. The equilibrium (undriven) horizon is the area endpoint, consistent with the area frameworks (Gibbons–Hawking, the Type II_1 maximum [52], DSSYK) as descriptions of the *eternal* horizon. Within the present paper the *value* $\Delta = 1$ is set by §5 (iii)'s birth-selection, which lands the grown cosmic horizon on the volume branch as a point; the geometric reframing ($\Delta = d_\mathrm{H} - 2$) and the ceiling $\Delta \leq 1$ are what the count question reduces to, now carried out in the count paper (the ceiling as a model-conditional theorem; the value $\Delta = 1$ as the stated $\mathrm{L} \to \infty$ endpoint choice). We do not resolve de Sitter holography here: the sharpest in-paper statement of what a resolution requires remains the II_1 trace calibration of the FRW patch, which §7.1b shows *is* the Δ measurement asked algebraically. The empirical Δ (DESI DR3 + Euclid, §7) is the arbiter.

8 Discussion

The chain reformulates SEDE's foundational input — and, in its present form, earns back a *qualified partial reduction.* The residual list has contracted to: the tensor-network description of the horizon state (the $S_\mathrm{ent} = S_\mathrm{grav}$ tier restated micro-structurally, §5 ii — the two-state dof is now the derived Ising dual of that description); the Planck-era clock normalisation behind the birth-selection probability (quantified, not assumed: $P(volume) \geq 0.9$, rising to 1 in the equilibration limit, volume-biased in every regime); and one scoped coefficient (the transverse diffusivity of the height equation, whose sink rate and source are now exact GR results, $\kappa = -H(1 - 3w)/4$ and $\delta R_\mathrm{A}/R_\mathrm{A} = -\delta M/2M$). The former "open dS-holography count" is no longer on the list as an independent item: the algebra type is fixed by the infrared attractor, and the ceiling's value is a trace calibration — the count *is* the Δ measurement, asked algebraically (§7.1b; and the standalone trace-calibration note). Assuming the tensor-network tier, the selection is a single causal chain from one premise, gravity's long-rangedness: exact mean-field thermodynamics sources the two-phase F(m) term-by-term (§5 ii); the N-dependent coupling $J/J_\mathrm{c} = 2\pi G m^2 N$ forces the ordering bifurcation at Planckian size, where Barrow's monotone slope (§5 i) tilts the nascent well — so the grown cosmic horizon tracks onto the volume branch, $\Delta = 1$ selected at birth with probability ≥ 0.9 ($\to 1$ in the equilibration limit), with no barrier ever crossed (§5 iii); and exact long-range hysteresis locks every horizon to its birth branch, making the black hole's $\Delta_{\rm BH} = 0$ a derived prediction rather than an accommodation. No z* near-critical transient is claimed (§6): the companion count paper [49] shows the apparent horizon is constraint-slaved (its eq. 2.9) and does not roughen, so the SOC/ϵ-lag layer that §5 (iv) might suggest does not survive — the exact membrane coefficients of §5 (iv) remain as GR facts, but they source no critical observable (§6). The genuinely established *reductions* in the chain are thermodynamic: ensemble inequivalence for the long-range de Sitter horizon ($C = -2S < 0$) removes the canonical area-law branch (§4), and the gate $f_{\rm sat}$ is derived from one-line activation kinetics (§7.1a). The reformulation yields falsifiability where the postulate had none: Δ is pinned to the pre-registered point $\Delta = 1$ (§5 iii; the ceiling $\Delta \leq 1$ is a theorem in the companion count paper [49], §7.1; frozen with an amendment protocol in the pre-registration), measured to $\sim$0.09 by DESI DR3 + Euclid (shared with the cosmology paper), with any intermediate value falsifying — the count paper makes the count discrete ($\Delta \in 0, 1$), so there is no intermediate value to accommodate. And the mechanism is already tested on public data: the gate rescues $\Delta = 1$ by $\Delta\chi^2 \approx 2400$ on DESI DR2, with the gated family preferring $\Delta = 0.93$ [0.83, 1.02] at zero dark-sector parameters (a profile likelihood, §7, Level 3). The residue is a clean hand-off — now in its sharpest form a *measurement*, not an open theory question — and the cosmology stands independently (the cosmology paper).

8.1 The covariant completion and the amplitude (status update)

Two clarifications supersede earlier framings of this programme and should be read as governing. **(i) The completion is local, and the parity argument is retracted.** The claim that an H-linear density $\rho_X \propto H$ *cannot* descend from a local conservative action is **false**: Kinetic Gravity Braiding (Deffayet–Pujolàs–Sawicki–Vikman 2010, arXiv:1008.0048) is an explicit local counterexample, and the completed theory is a finite cubic-KGB action $\mathcal{L} = M_{\rm P}^2 R/2 + K(X, \phi) - G_3(X, \phi)\Box\phi$ (FPAB-SEDE) whose linear perturbations are ghost/gradient-stable with $c_T = 1$ (the cosmology paper §3.4). The former time-reversal "parity no-go", the unique-$\Delta = 1$ *selection* claim, and the reading of $\rho_{\rm DE} = T\,s$ as a dark-sector *identity* are accordingly withdrawn: $\rho_X \propto H$ and $\Delta = 1$ are the phenomenological branch being completed, not consequences of a parity theorem, and the birth-selection of §5 is a *motivated conditional*, not a derivation (§8 above). Israel–Stewart bulk viscosity does **not** establish perturbative stability across the crossing (the single-fluid principal symbol diverges at $\rho + p = 0$); braiding does. **(ii) The amplitude is not derived.** The scale $\mu \approx 28.5$ MeV is the coincidence problem in dimension-three form, shared with Λ; the counting/network construction motivates the *form* $\Delta = 1$ but does not fix μ. A numerical relation $\mu^3 \simeq \alpha_{B0}\, f_\pi^2 m_\pi^2 / m_{\eta'}$ reproduces the closure value to 1.01% (predicting the braiding fraction $b \approx 0.206$); we present this as a **coefficient-selection hypothesis**, not a Standard-Model derivation — the topological-coefficient identification is an ansatz and the structure gate is not obtained from QCD. The genuine microscopic home of the gate is the 2PI/closed-time-path effective action of a structure reservoir (reciprocal matter–scalar stress exchange constrained by the Bianchi identity); its kernel is not derived and it is labelled an extension, not the established core. **(iii) The dissipation is driven, not thermal — an FDT/KMS retraction.** An earlier companion derivation presented the dissipative dark-energy dynamics as a dynamical-KMS structure at the horizon temperature, with $\zeta > 0$ identified with the fluctuation–dissipation lock $\langle \xi\xi \rangle = 2\zeta T_{AH}$. That claim is retracted, because the calculation was done: the noise that actually drives the gate is cold and structure-sourced — set by the primordial amplitude A_s, not by the 10^{-33} eV horizon temperature — and exceeds the KMS-thermal value by $\sim 10^{57}$. The *mean* horizon thermodynamics of the background ($T_{AH} = H/2\pi$, the H-linear form) is untouched, but the correct framing of the dissipative sector is the driven, entropy-producing NESS used throughout this paper, with $\zeta > 0$ surviving as the second law — not a thermal-KMS effective field theory. We record the correction because it is itself informative: it unifies the dawn and dusk sectors of the wider programme as non-equilibrium at both ends. **(iv) The tensor sector is exactly GR here too — $\delta(z) = 0$, $\Xi_0 = 1$.** The driven-NESS framing reproduces the cosmology paper's tensor result, so the standard-siren prediction $\Xi_0 = 1$ is a joint statement of both papers. Four channels close it. *(a) No new bulk fields:* the dynamical content — the horizon order parameter, the hysteresis lock, the structure reservoir — lives in the horizon *accounting* and the scalar/entropy sector; the covariant bulk completion is FPAB, whose tensor sector is exactly GR ($\alpha_M = \alpha_T = 0$, cosmology paper §6). The conclusion depends only on *where* the new physics lives, so it survives the open items in the §5 mechanism. *(b) Dissipative bulk channel, bounded to irrelevance:* the most uncharitable reading endows the driven medium with an effective shear viscosity damping GWs at the Hawking rate $\Gamma = 16\pi G\eta/c^2$; an $\mathcal{O}(1)$ propagation effect over a Hubble time needs $\eta \approx 6 \times 10^7\ Pa \cdot s$, whereas a deliberately generous kinetic estimate of the cosmic medium ($\rho_m\, \sigma_v \times 1$ Mpc) gives $\approx 24\ Pa \cdot s$, so $|\Xi_0 - 1| \lesssim 4 \times 10^{-7}$ — five orders below the forecast siren sensitivity $\sigma(\Xi_0) \sim 10^{-2}$, and itself conservative (LIGO-band wavelengths are far below any structure scale, so the fluid regime does not even apply). *(c) Horizon-boundary absorption, closed by our own result:* the apparent-horizon condition $\theta_+ = 0$ is elliptic (constraint-slaved, no growth equation; count paper eq. 2.9) — a boundary with no dynamical modes cannot resonantly absorb the sub-horizon tensor radiation ($k/\mathcal{H} \sim 10^{18\text{–}20}$ in the LIGO band). *(d) Fluctuation–dissipation consistency:* zero tensor dissipation pairs with zero horizon-sourced tensor noise ($\Omega_{\rm gw} \sim (H/M_{\rm P})^2 \sim 10^{-122}$). This is consistent with clarification (iii): the *scalar* gate is a cold, structure-driven NESS, while the *tensor* sector sees an equilibrium GR vacuum channel in both formulations. One conditional remains, at low risk: if the reservoir kernel is ever realised as a graviton-mediated transport channel it would add an utterly negligible *source* to the stochastic background, not a propagation modification, so $\Xi_0 = 1$ survives even that branch.

For honesty the branch selection itself is **gravitationally silent**: the ordering bifurcation of §5 is a horizon-ledger transition with no bulk latent heat, no bubble nucleation and no propagating horizon modes, so it emits no gravitational-wave signature — its only observable trace is the $\Delta = 1$ it leaves behind. Relatedly, $\Delta_{\rm BH} = 0$ (black holes locked to the area law, §5, §8) means SEDE predicts exactly standard Bekenstein–Hawking black-hole thermodynamics: unmodified Kerr quasinormal spectra and no echoes. This separates SEDE from quantum-corrected-entropy models that predict near-horizon structure — a confirmed ringdown anomaly attributable to modified black-hole entropy would falsify SEDE, and the continuing null echo searches are (weak) SEDE-consistent evidence.

8.2 Reproducibility

Every quantitative claim in this paper is reproduced by the accompanying code from a single entry point, each stage carrying validation assertions; all stages pass. The four figures are regenerated from the same code.

9 Conclusion

We set out to test how much of SEDE's single foundational input — the volume-law (Barrow $\Delta = 1$) horizon entropy — is irreducible. Decomposed into *state, form, scale,* and *count,* the first three reduce to first principles given the standard entanglement identification of the horizon entropy, and a Bekenstein no-go shows the residual count cannot follow from any energy bound. What replaces the postulate is a *selection*: grown continuously from Planckian size, the cosmic horizon passes through the ordering bifurcation of a driven long-range system with Barrow entropy tilting the nascent well toward volume, so $\Delta = 1$ is selected at birth — with probability ≥ 0.9, rising to unity in the equilibration limit — and long-range hysteresis locks each horizon to its birth branch: the cosmic horizon to volume, an abruptly-formed black hole to area.

The significance is a change of epistemic status rather than a claimed proof. An untestable counting postulate becomes a falsifiable prediction: Δ is pinned to the pre-registered point $\Delta = 1$ (§5 iii; the ceiling $\Delta \leq 1$ is a theorem in the companion count paper [49]), decided to $\sigma(\Delta) \approx 0.09$ by DESI DR3 + Euclid, with any intermediate value falsifying — the count paper makes the count discrete ($\Delta \in 0, 1$). On present public data the mechanism is already evident — the structure-gated family prefers the volume endpoint at zero dark-sector parameters, which the un-gated Barrow family pays $\Delta\chi^2 \approx 2400$ to reach. The residue is thus handed off as a *measurement*, not an open theory question. The one premise still to be earned is the tensor-network description of the horizon state, on which the selection chain is conditional; future work should target that description directly. The cosmological results of the cosmology paper stand independently of the reduction attempted here.

Data availability statement

All code and analysis pipelines are publicly available at `https://github.com/spsingularity/sede-foundations`. Every quantitative claim in this paper is produced by the accompanying code wired into a single entry point, each stage carrying validation assertions; the figures are regenerated from the same code. The model and its cosmology live with the companion cosmology paper [5]; no additional observational data beyond the cosmology paper's standard public inputs is used here. A tagged release is archived at Zenodo, DOI `[Zenodo DOI | assigned on release]`.

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